

Systematic Time-Window, Delay, and Rank Verification for Local DMD of `mouse02_sleep`

Reproducible development-only analysis report

17 July 2026

Abstract

This report determines what time window and calibration coefficient were used in the preceding analysis, then tests a standard local dynamic mode decomposition (DMD) tuning pipeline on one NREM recording from `mouse02_sleep`. The preceding code selected a 196.08-s raw block but fitted DMD to only 117.65 s. The corrected analysis uses PyDMD 2025.8.1, four-frame event-rate bins, a fixed acquisition-1 observation basis, common chronological forecast origins, and a joint scan over 20–180-s fitting windows, five delay depths, and seven explicit ranks. Two prediction coefficients are kept deliberately separate. The primary R^2_{fixed} uses the same acquisition-1 mean reference for every window and therefore permits a fair comparison across W ; the auxiliary R^2_{local} measures only the increment beyond each window’s own past mean.

No guarded configuration achieved positive R^2_{fixed} , and none passed the technical development gate. The lowest comparable loss occurred at $W = 180$ s, $d = 1$, $r = 12$, with $R^2_{\text{fixed}} = -0.004833$ and $R^2_{\text{local}} = 0.004651$. Because there was no gate-passing pool, a one-standard-error parsimony calculation produced only a *fallback*: $W = 180$ s, $d = 1$, $r = 2$, with $R^2_{\text{fixed}} = -0.005171$ and $R^2_{\text{local}} = 0.004224$. The 45-s, rank-4 model has the largest local increment, $R^2_{\text{local}} = 0.008932$, but its comparable score is $R^2_{\text{fixed}} = -0.042152$. Thus the fair forecast screen favors longer windows only in the limited sense that they lose less; it does not verify a 180-s model.

A damping-aware spectral audit found no eligible oscillatory pair in either the 45-s local-increment candidate or the 180-s fallback. Longer delayed models sometimes produced eligible signatures, but the available origins were too overlapping to verify recurrence. A separately refitted, development-internal Residual-DMD-style diagnostic gave a globally synchronized surrogate $p = 0.13$, unadjusted for candidate selection. Training reconstructions explained essentially none of the all-neuron variance. All optimized-DMD fits emitted convergence warnings. The outer 41.83-s tail consequently remains score-locked and unscored. Neither 45 s nor 180 s is a verified window for brain-state mode tracking.

1 Direct answers and interpretation

1.1 What was the previous DMD window?

Three durations had been conflated. The selected raw block contained 1500 native frames, or

$$\frac{1500}{7.65} = 196.08 \text{ s.}$$

Four-frame binning produced 375 samples. The PyDMD fit used only the first 225 bins,

$$225 \frac{4}{7.65} = 117.65 \text{ s.}$$

The next and final 75-bin blocks were each 39.22 s. Therefore 117.65 s, not 196.08 s, was the preceding DMD fitting duration.

1.2 What was calibration R^2 ?

The earlier quantity was a held-out one-step prediction coefficient pooled over neuron–time entries,

$$R_{\text{cal}}^2 = 1 - \frac{\sum_{i,t} (y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t} (y_{i,t} - \bar{x}_{i,\text{train}})^2}.$$

It was not probabilistic calibration, autonomous trajectory reconstruction, a Residual-DMD statistic, or evidence that a dynamic mode was stable. The old delay comparison also allowed the automatic SVD rank to change with delay, so it did not isolate delay depth. This report avoids the ambiguous term *calibration* R^2 and instead defines a comparable fixed-reference prediction coefficient and a separate local-reference increment.

1.3 Should the fitting window simply be shorter?

No general rule makes the shortest window preferable. A short window may reduce mixing between changing dynamical regimes, but it also gives fewer regression transitions, a noisier local mean, weaker rank support, and poorer resolution of slow oscillations. A long window reverses those advantages and disadvantages. For N_W binned samples and delay d , the number of snapshot transitions is $N_W - d$, the explicit delay memory is $(d - 1)\Delta t$, and the delayed ambient dimension is qd .

The present data do not support the claim that a short window is better. The fair cross-window loss decreases toward 180 s, whereas the small improvement relative to a window-specific local mean peaks at 45 s. Every comparable score nevertheless remains negative. Online and windowed DMD likewise treat locality as a parameter to validate rather than a universal optimum [4, 5]. The scientifically honest conclusion is therefore not that 180 s works, but that no tested window currently works.

2 Data, observable, and chronological design

2.1 One dataset and one state

Only `mouse02_sleep.mat` and raw state code 1 (NREM) were used. No paper-specific `used_frame` mask entered this analysis. Every bin, delay vector, fitting window, and target remained within one NREM run and one acquisition. This state restriction prevents an explicit scored state transition inside a window, but it does not prove that a single stationary dynamical system generated the complete NREM interval.

Four native deconvolved-event samples were summed and divided by

$$\Delta t = \frac{4}{7.65} = 0.522876 \text{ s}.$$

The resulting observable is event mass per second. It is a continuous-valued rate proxy constructed from event data, not a calibrated spike rate.

2.2 Fixed neuron transformation and observation basis

Acquisition-1 NREM bouts supplied the fixed neuron transformation. Of 6,574 neurons, 6,570 had finite nonzero acquisition-1 RMS. For eligible neuron i ,

$$x_i(t) = \frac{r_i(t) - \mu_{i,\text{acq1}}}{\text{RMS}_{i,\text{acq1}} + 10^{-3}}.$$

A POD basis B_q was learned only from these standardized acquisition-1 NREM samples. All development targets came from acquisition 2. For a candidate fitting window, the DMD coordinates were

$$z(t) = B_q^\top \{x(t) - \bar{x}_W\}, \quad \hat{x}(t+h) = B_q \hat{z}(t+h) + \bar{x}_W,$$

where $\bar{x}_W$ is computed from the past fitting window only. Forecast loss was evaluated in the complete 6,570-neuron norm. An exact orthonormal-projection identity retained error outside $\text{col}(B_q)$, so the sweep did not silently score only the retained POD coordinates.

2.3 Development and outer geometry

The longest acquisition-2 NREM bout contained 516 complete rate bins, or 269.80 s. Eight common development origins were separated by 12 bins; their eight-bin future target blocks did not overlap, although their fitting contexts did. The final 80 bins, or 41.83 s, were assigned to five possible outer origins. The corrected run defined this outer interval by time and state label but did not display or score its neural activity.

An earlier exploratory draft had displayed that activity. The current report therefore calls the interval *score-locked and unscored*, not pristine or untouched. It is not opened unless the complete development gate passes.

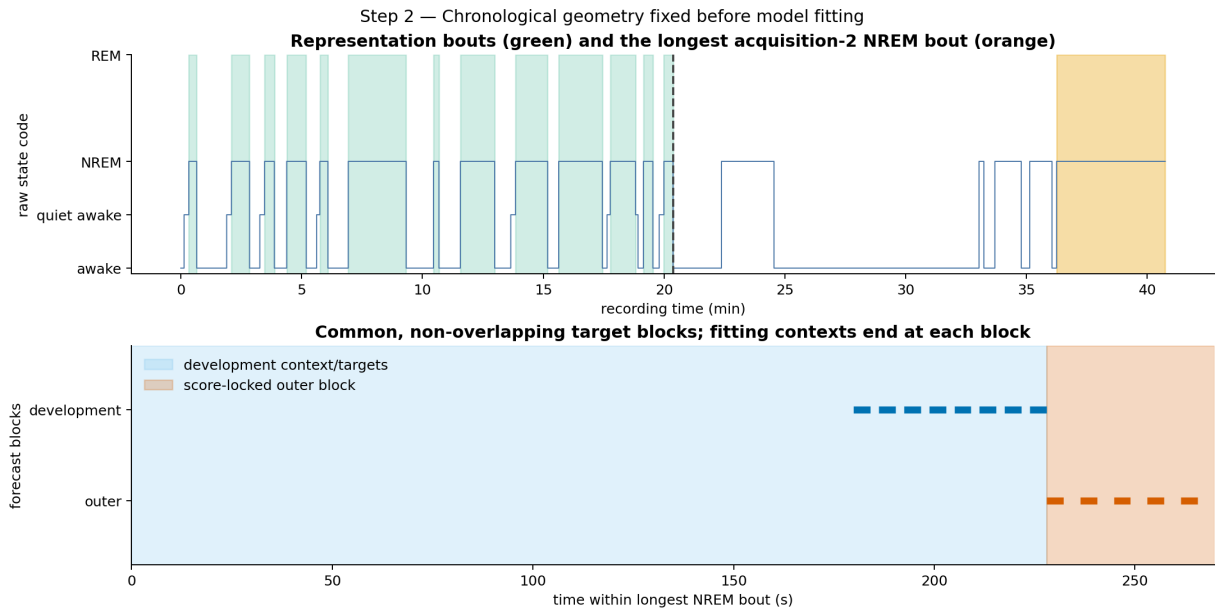


Figure 1: Acquisition-1 representation bouts and the acquisition-2 development/outer geometry. The orange outer interval is defined, but its activity and scores remain closed.

3 Two prediction references that answer different questions

Let o index a forecast origin and $\mathcal{H} = \{1, 2, 4, 8\}$ the scored horizons in rate bins. The primary loss is the equal-weight mean

$$\mathcal{L}_{\text{fixed}} = \text{mean}_{o, h \in \mathcal{H}} \frac{\text{SSE}_{\text{DMD}}(o, h)}{\text{SSE}_{\mu_{\text{acq1}}}(o, h)}, \quad R_{\text{fixed}}^2 = 1 - \mathcal{L}_{\text{fixed}}.$$

After acquisition-1 standardization, $\mu_{\text{acq1}} = 0$. At a given origin and horizon the denominator is identical for every W, d, r , which makes R_{fixed}^2 the appropriate cross-window selection metric.

The auxiliary local increment is

$$\mathcal{L}_{\text{local}} = \text{mean}_{o, h \in \mathcal{H}} \frac{\text{SSE}_{\text{DMD}}(o, h)}{\text{SSE}_{\bar{x}_W}(o, h)}, \quad R_{\text{local}}^2 = 1 - \mathcal{L}_{\text{local}}.$$

It asks whether the autonomous DMD forecast improves on the same window’s past-only constant prediction. This is scientifically useful, but its denominator changes with W , so it cannot fairly select a window.

Table 1: Mean denominator over the common development origins and horizons. The fixed denominator is constant across W ; the local denominator is not.

W (s)	fixed-reference SSE	local-mean SSE
20	10880.72	11375.86
30	10880.72	11238.00
45	10880.72	11112.39
60	10880.72	11026.05
90	10880.72	10951.64
120	10880.72	10914.67
180	10880.72	10892.08

The short-window local mean is, on average, worse than the fixed acquisition-1 mean. A model can consequently have a small positive R_{local}^2 while remaining negative under R_{fixed}^2 . That is exactly what occurs here. Reporting only R_{local}^2 would make the 45-s peak look like a cross-window optimum when it is not.

4 Representation-capacity audit

The original $q = 64$ POD basis retained only 22.03% of standardized acquisition-1 energy. Before fitting DMD, the corrected analysis computed an instantaneous projection oracle on the development targets. For each future truth, the oracle used the best possible prediction in $\bar{x}_W + \text{col}(B_q)$. It is future-informed and is therefore not forecast performance. It asks only whether the observation space makes the declared effect-size target attainable.

The 64- and 128-dimensional representations could not reach the declared $R^2 = 0.05$ target even with future information. The corrected run fixed $q = 256$, the smallest tested dimension that made the target attainable. The practical effect target was prespecified on R_{local}^2 ; the fixed oracle is additionally reported because R_{fixed}^2 controls cross-window selection. Every fitted score was checked programmatically to remain below the corresponding oracle ceiling.

Table 2: Nested POD capacity. The two oracle columns are maxima over the tested windows and use the indicated prediction reference.

q	retained energy	max fixed oracle R^2	max local oracle R^2
64	22.03%	0.02022	0.03620
128	34.76%	0.03155	0.04739
256	52.74%	0.05260	0.06811

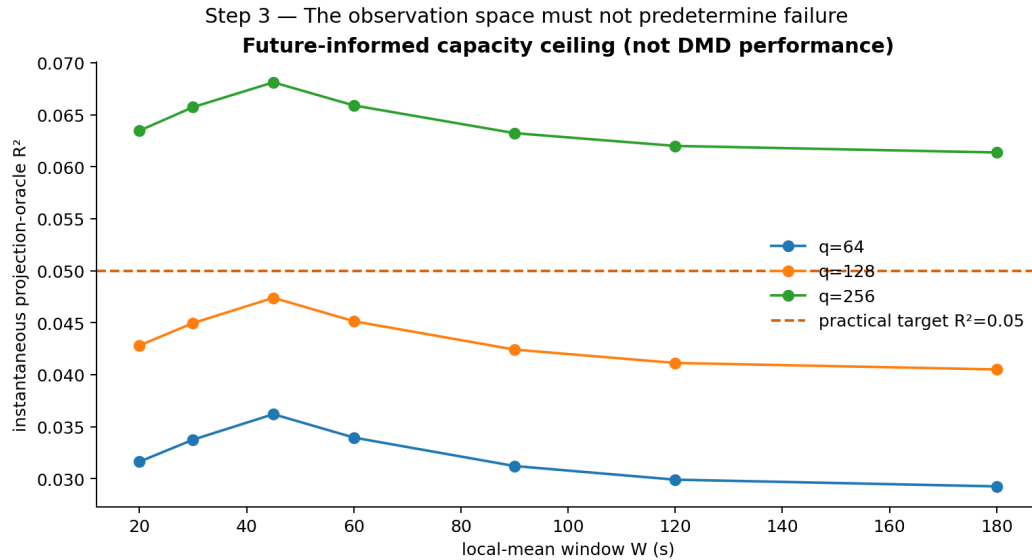


Figure 2: Future-informed development projection ceilings. These curves measure observation-space capacity, not DMD prediction.

5 Locked window, delay, and rank tuning pipeline

5.1 Candidate grid and support guard

The candidate grid was

$$W \in \{20, 30, 45, 60, 90, 120, 180\} \text{ s}, \quad d \in \{1, 2, 4, 8, 16\},$$

$$r \in \{2, 4, 8, 12, 16, 24, 32\}.$$

Rounded fitting durations were 19.869, 29.804, 44.967, 60.131, 89.935, 120.261, and 179.869 s. Delay memories were 0, 0.523, 1.569, 3.660, and 7.843 s. A candidate entered selection only if

$$\frac{N_W - d}{r} \geq 8.$$

This eight-transitions-per-rank requirement is a transparent project heuristic, not a DMD theorem. It retained 143 of 245 displayed combinations.

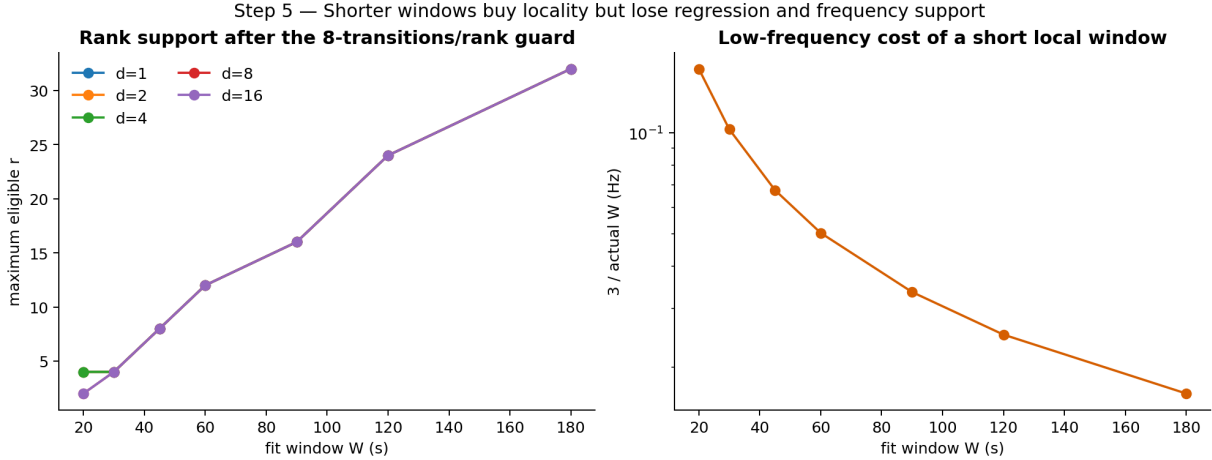


Figure 3: Regression support, delay memory, and rough frequency-resolution trade-offs over the locked grid.

5.2 Primary estimator and autonomous scoring

The primary estimator was PyDMD’s `HankelDMD` with explicit `svd_rank`, `exact=True`, and the candidate delay d . TLSQ rank was zero; optimized amplitudes, forward-backward correction, rescaling, and Tikhonov regularization were disabled. PyDMD exposes these as separate choices [2]; the primary screen changed only W, d, r so their effects remained interpretable.

At each origin the complete past window was fitted once. The last observed delay state initialized one autonomous eight-bin forecast. No future observation was inserted after initialization. Persistence and a stable neuronwise AR(1) were additional dynamic baselines. The implementation also contains a noise-free autonomous-forecast unit check and an exact all-neuron POD-score identity check.

5.3 Technical and practical gates

A technically successful configuration required:

1. positive mean R_{fixed}^2 and R_{local}^2 ;
2. positive fixed and local scores separately at $h = 1$ and $h = 8$;
3. positive-origin fractions of at least 0.75 for both references overall and at the endpoint horizons;
4. positive mean skill against persistence and AR(1), with at least 0.75 of origins beating both;
5. finite numerical diagnostics and no unexpected warning.

The practical gate additionally required $R_{\text{local}}^2 \geq 0.05$, stable selection, recurrent modes, and the separate residual diagnostic. Repeated rolling origins and multiple horizons follow general out-of-sample forecast evaluation practice [6]; the numerical thresholds remain project heuristics.

A delete-two-origin jackknife supplied a descriptive one-standard-error parsimony calculation. The fitting contexts overlap, so this is not a formal independent-sample standard error. Most importantly, when no configuration passes the technical gate, its output is labelled a *fallback*, not a selected or verified model.

6 Forecast results: longer loses less, but no window succeeds

All 1,144 guarded exact-Hankel development fits returned and none failed. There were zero technically passing configurations. All 143 configurations had negative R_{fixed}^2 , while 98 had positive R_{local}^2 . The latter counts incremental improvement over a window-dependent and often inferior local-mean baseline; it is not evidence that 98 models forecast usefully.

Table 3: Lowest comparable loss within each W . The table reports both references because they answer different questions.

W (s)	d	r	R_{fixed}^2	R_{local}^2	skill-persist.	skill-AR(1)
20	1	2	-0.084123	0.006792	0.4871	0.2026
30	1	4	-0.060204	0.007092	0.4965	0.1839
45	1	4	-0.042152	0.008932	0.5036	0.1604
60	1	12	-0.028138	0.007543	0.5087	0.1273
90	1	2	-0.015195	0.005500	0.5135	0.1114
120	4	2	-0.009526	0.004363	0.5155	0.0867
180	1	12	-0.004833	0.004651	0.5179	0.0287

The raw comparable minimum is $W = 180$ s, $d = 1$, $r = 12$. Its positive skill against persistence and AR(1) does not contradict its negative R_{fixed}^2 : both dynamic baselines are worse than the fixed acquisition mean. The rank-12 model is therefore the *least poor comparable forecast*, not a successful model.

Table 4: Development scores for the 180-s, $d = 1$, $r = 2$ fallback.

horizon (bins)	horizon (s)	R_{fixed}^2	R_{local}^2
1	0.523	0.000816	0.008923
2	1.046	-0.008724	0.002350
4	2.092	-0.010630	0.005676
8	4.183	-0.002145	-0.000051

No gate-passing pool existed. Applying the descriptive one-SE rule to the complete guarded grid produced $W = 180$ s, $d = 1$, $r = 2$. This result is recorded as **fallback_no_gate_passing_configuration**. Its horizon-specific scores are:

The fallback fails both the mean common-reference requirement and the eight-bin local-reference endpoint. It must not be called the final verified model.

6.1 Delay does not rescue short windows

Table 5: Best comparable result for each delay over all guarded W, r .

d	W (s)	r	R_{fixed}^2	R_{local}^2
1	180	12	-0.004833	0.004651
2	180	2	-0.006295	0.003090
4	180	12	-0.006835	0.002602
8	180	2	-0.007278	0.002230
16	180	2	-0.008596	0.000976

The hypothesized pattern that a short W needs a large d is not supported for this observable. Delay embedding may still be necessary for other observables and systems; PyDMD’s documented workflow recommends examining singular support and validation performance rather than assuming a delay depth [1].

6.2 Selection instability

Deleting consecutive origin pairs produced fallback one-SE tuples

$$(W, d, r) \in \{(120, 4, 2), (180, 1, 2), (180, 1, 2), (180, 1, 2)\}.$$

The full fallback recurred in three of four screens, but none of the four reduced screens had any gate-passing pool. The 0.75 tuple match rate is therefore not successful scientific selection stability.

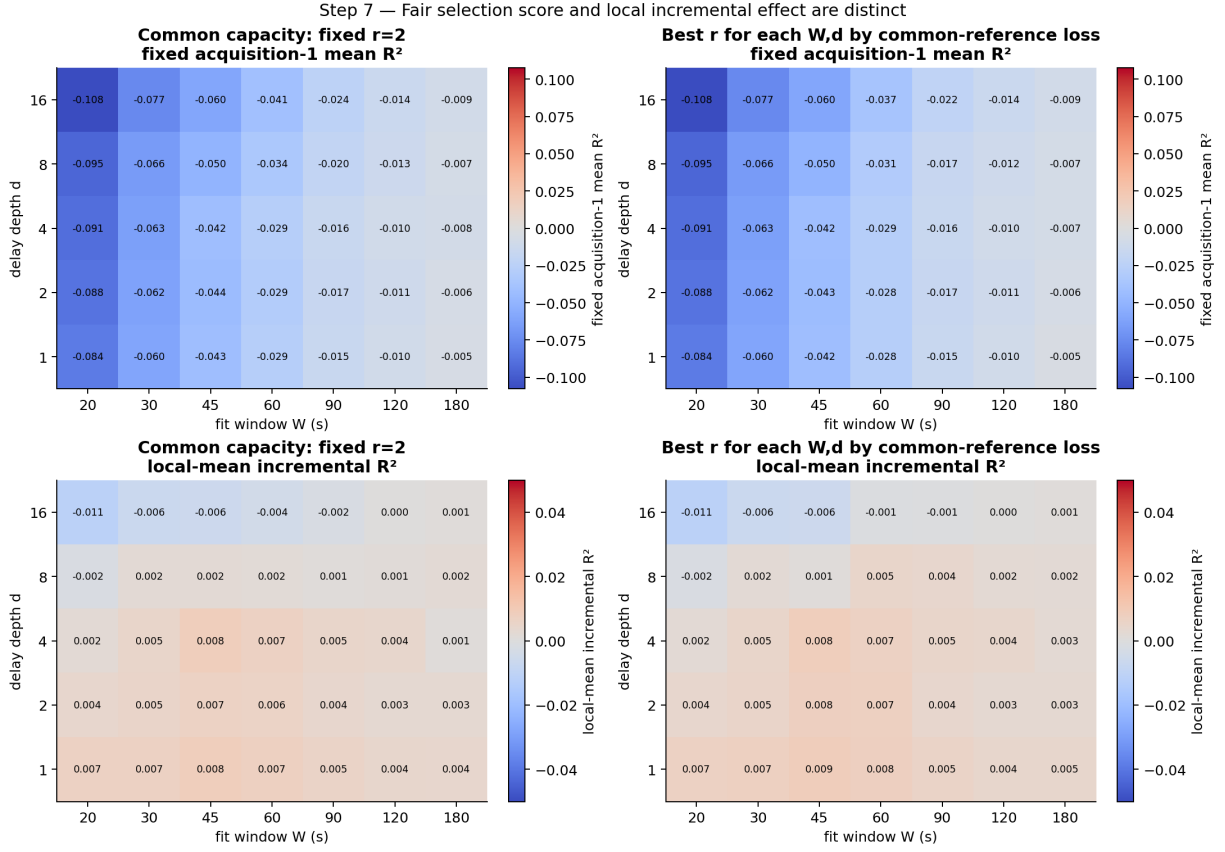


Figure 4: Fixed-reference and local-reference development scores. The fixed-reference panels permit comparison across W ; the local panels show only the increment beyond each window’s own past mean.

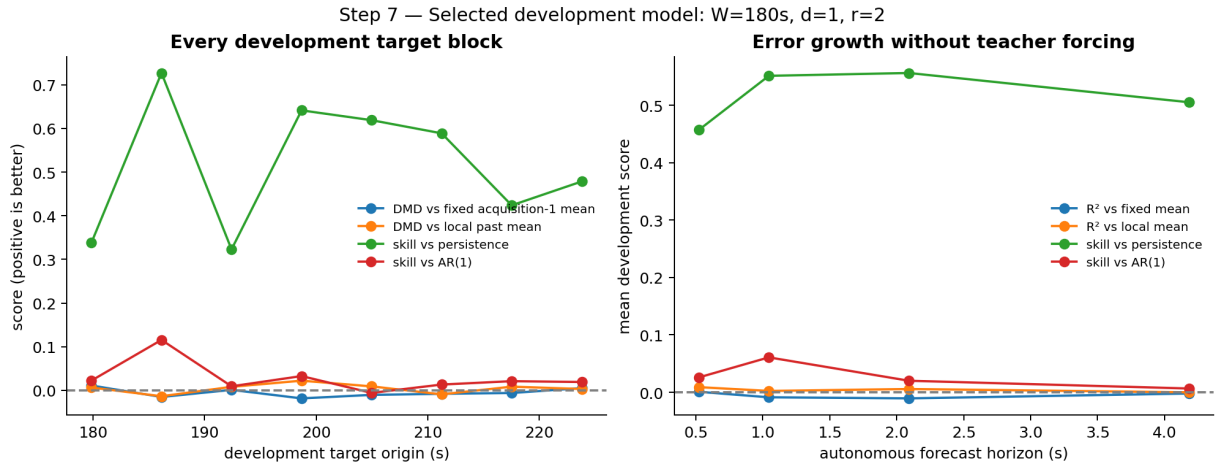


Figure 5: Origin and horizon diagnostics for the one-SE fallback. A displayed fallback is not a gate-passing selection.

7 Numerical and noise-aware checks

7.1 Exact-Hankel warnings and conditioning

Of 1,144 exact fits, 864 emitted PyDMD’s input-condition warning. All were classified as expected rank-truncation warnings and none was unexpected. The warnings occurred for the shorter matrices; the 180-s finalists emitted none. No exact fit failed.

For the 180-s, $d = 1, r = 12$ comparable minimum, the median retained condition number σ_1/σ_{12} was 4.171, while the median boundary ratio σ_{12}/σ_{13} was only 1.016. For the rank-2 fallback, the corresponding medians were 1.909 and 1.300. Full operator-pair and centered input condition numbers were approximately 99. The explicit truncations are numerically manageable, but the weak rank-12 boundary does not support interpreting 12 as a sharply identified intrinsic dimension.

7.2 Optimized/BOP-DMD sensitivity

PyDMD’s noisy-data tutorial demonstrates delay coordinates with optimized and bagged optimized DMD [1, 3]. A locked shortlist therefore tested optimized DMD with conjugate-pair constraints, with and without an additional stability constraint. In PyDMD, `num_trials=0` invokes optimized DMD without bootstrap bagging.

There were 16 estimator/configuration variants and 128 origin-level fits. Every origin fit emitted a convergence warning. Their returned objective values are not treated as valid performance estimates, and no convergence-clean optimized candidate was available to promote. Variable projection can be sensitive to nonlinear initialization and local solutions [7]; a returned object alone is not evidence of successful optimization.

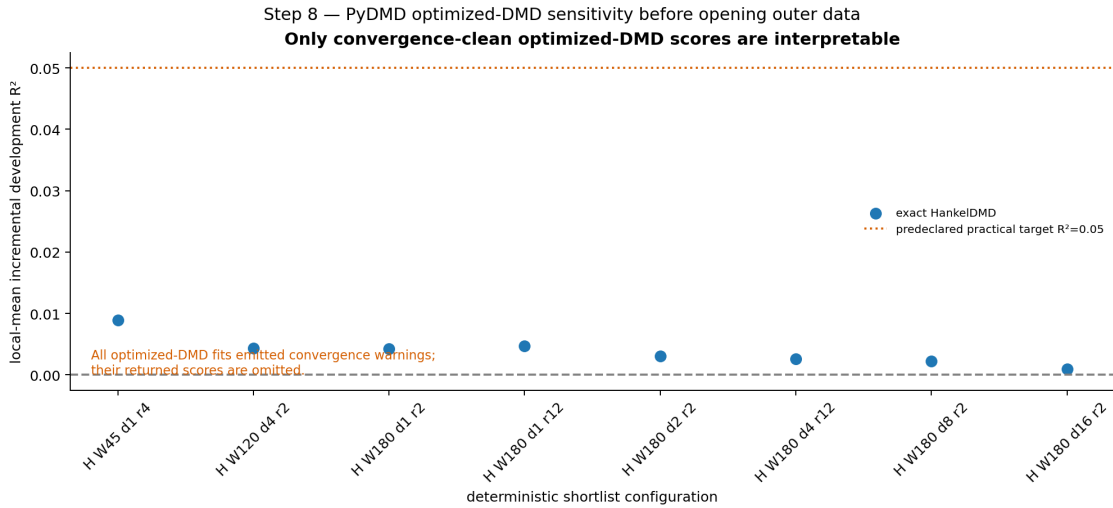


Figure 6: Exact-Hankel shortlist performance. Nonconverged optimized-DMD scores are deliberately omitted from interpretation.

7.3 Raw-rate inner-product sensitivity

Neuronwise RMS scaling gives low- and high-rate neurons similar weight. The complete grid was therefore repeated in raw-centered event-rate units with a separate fixed $q = 256$ basis, which retained 58.36% of acquisition-1 raw-rate energy.

The fair raw-rate minimum was $W = 180$ s, $d = 1$, $r = 4$, with $R_{\text{fixed}}^2 = -0.003086$ and $R_{\text{local}}^2 = 0.005302$. The largest raw local increment was again at $W = 45$ s, $d = 1$, $r = 4$: $R_{\text{local}}^2 = 0.010190$, but $R_{\text{fixed}}^2 = -0.029063$. No raw-rate configuration passed. The sensitivity analysis reproduces the distinction between a long-window comparable minimum and a small 45-s local increment; it does not validate either model.

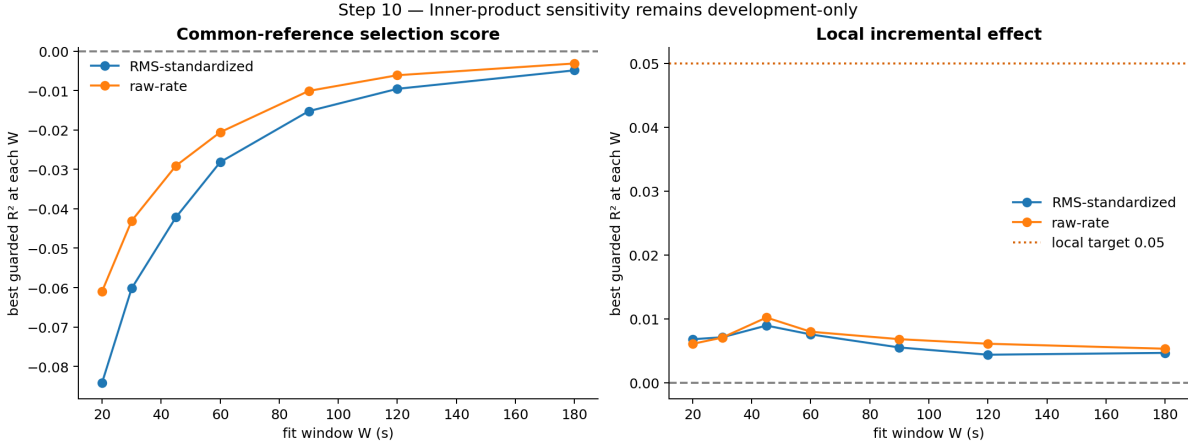


Figure 7: Raw-rate and standardized scoring. Comparable fixed-reference selection and window-specific local increments are shown separately.

8 Within-window locality is still an approximation

State-label homogeneity prevents an explicit NREM-to-other-state transition inside a scored window; it does not establish stationarity. A descriptive diagnostic split the first eight fixed POD coordinates into chronological halves and compared their means, covariances, ridge-stabilized one-step operators, and cross-half one-step transfer.

Table 6: Descriptive split-half locality diagnostics.

W (s)	mean shift	covariance drift	operator drift	transfer R^2	positive fraction
20	0.646	1.439	1.378	-1.8885	0.000
30	0.558	1.189	1.306	-0.2696	0.625
45	0.471	0.983	1.207	0.1805	0.750
60	0.429	0.933	1.081	0.3917	1.000
90	0.300	0.601	1.089	0.5133	1.000
120	0.343	0.577	0.941	0.5501	1.000
180	0.187	0.318	0.776	0.5627	1.000

These values do not show that short windows are more stationary. They also do not certify the 180-s window: a longer half contains more fitting samples, so improved transfer can reflect reduced estimator variance rather than a more constant underlying system. The diagnostic is not a gate in this pilot. A confirmatory study should prespecify state-preserving block surrogates or change-point exclusions before using drift as a decision rule.

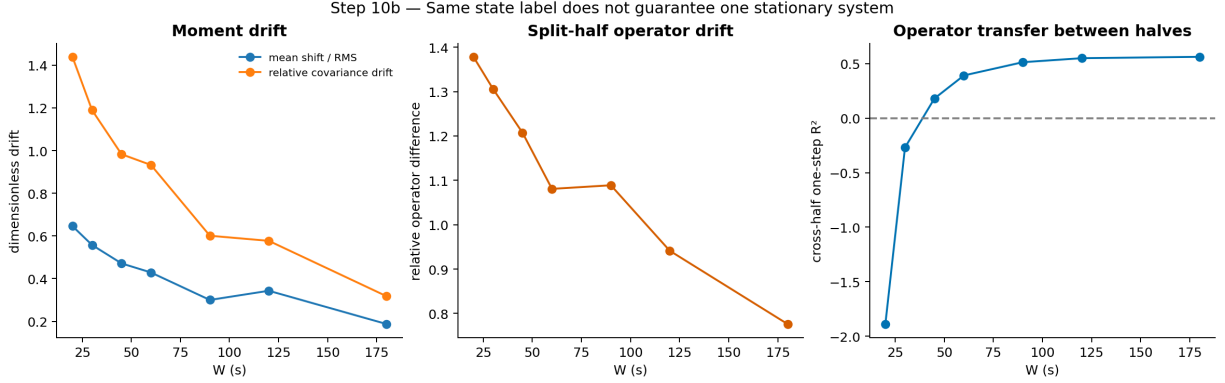


Figure 8: Split-half locality diagnostics in a fixed eight-dimensional space. They expose, but do not resolve, the stationarity approximation.

9 Damping-aware mode recurrence

9.1 Why frequency alone is insufficient

Forecast success and spectral adequacy are separate requirements. A frequency-resolved eigenvalue can decay so quickly that it does not describe a recurrent oscillatory component. For a positive-frequency conjugate pair, the audit therefore used the actual evidence span

$$T_{\text{evidence}} = (N_W - d)\Delta t$$

and required

$$fT_{\text{evidence}} \geq 3$$

cycles. It also required at least 1% of a descriptive fitted-component Frobenius-power proxy. This proxy uses per-snapshot least-squares coefficients; it is not physical mode energy.

For continuous rate

$$\omega = \frac{\log \lambda}{\Delta t},$$

the retained amplitude after one oscillatory cycle is

$$m_{\text{cycle}} = \exp\left(\frac{\text{Re } \omega}{f}\right).$$

The pilot persistence threshold was $m_{\text{cycle}} \geq 0.10$: at least 10% of the amplitude must survive one period. This transparent threshold is a project heuristic, not a DMD theorem.

Eligible pairs were matched across origins when their frequencies differed by no more than one Rayleigh bin, the affinity between their real spatial mode planes was at least 0.70, and their log

per-cycle multipliers differed by at most one. A recurrent track had to cover at least six of eight origins and directly match at least 75% of at least two available origin pairs whose fitting contexts shared no more than 25% of their samples.

9.2 Observed mode inventory

Across the deterministic exact-Hankel shortlist there were 179 positive-frequency candidate pairs. Of these, 160 completed at least three cycles, 173 passed the 1% proxy threshold, 46 passed the damping threshold, and 45 passed all eligibility conditions. The median per-cycle multiplier over all candidate pairs was only 0.01006.

The candidate-specific results are decisive:

- The 45-s, $d = 1, r = 4$ local-increment candidate produced nine positive-frequency pairs across eight origins. Eight were frequency-resolved and all nine were salient, but none retained 10% amplitude for one cycle. The median multiplier was 0.02929 and the maximum was 0.03984.
- The 180-s, $d = 1, r = 12$ comparable minimum produced 38 positive-frequency pairs. None passed the damping threshold; the median multiplier was 1.69×10^{-5} and the maximum was 0.03687.
- The 180-s, $d = 1, r = 2$ fallback produced no positive-frequency conjugate pair.

Some delayed 60- and 180-s models did produce damping-qualified signatures. However, every $W \geq 60$ candidate had fewer than two sufficiently separated origin pairs in this bout. Adjacent fitting contexts overlapped by 89.57% at 60 s and 96.51% at 180 s. The 45-s geometry offered three low-overlap origin pairs, but that candidate had no damping-qualified mode. Consequently no exact-Hankel mode track was verified. Long-window recurrence is *geometrically inconclusive*; it is not evidence of recurrence.

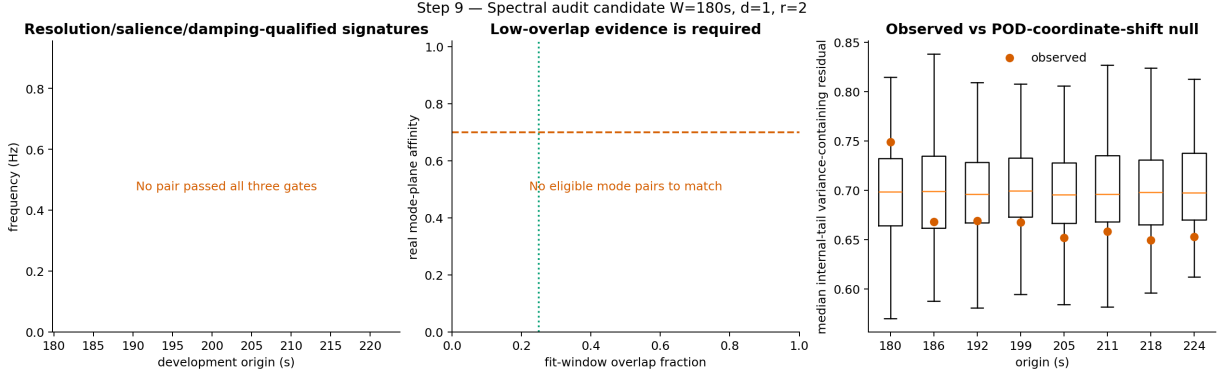


Figure 9: Damping-qualified frequency signatures, spatial-plane matching versus fitting-context overlap, and the separate internal Residual-DMD-style diagnostic.

Step 9 — All-neuron audit-candidate modes at the last development origin; no neuron sampling

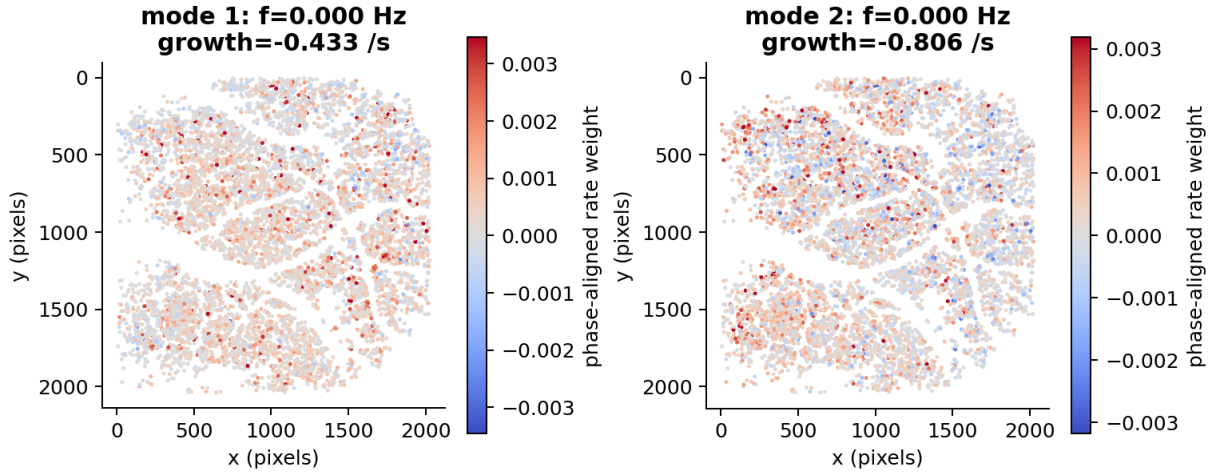


Figure 10: Phase-aligned all-neuron maps for the audit candidate at the last development origin. They are descriptive fitted maps; no oscillatory mode from the fallback passed the spectral gate.

10 Residual-DMD-style diagnostic with a synchronized null

PyDMD has no Residual-DMD fitter; its `RDMD` class denotes Randomized DMD. The present calculation therefore refitted a separate rank- r EDMD dictionary to the first part of each context and evaluated the published G, A, L residual quadratic form on an internal chronological tail [8]. It is not a residual attached to the displayed full-window PyDMD modes.

Because no spectrally qualified finalist existed, the diagnostic audited the development fallback $W = 180$ s, $d = 1$, $r = 2$. Each 344-bin context used 258 bins (134.90 s) for dictionary fitting and 86 bins (44.97 s) for quadrature. The centering mean and dictionary were learned from the fit prefix only. These tails had nevertheless already participated in model development, so they are not post-selection confirmation.

The null used 199 fixed-POD-coordinate circular-shift surrogates. Each surrogate was generated once over the complete accessible development prefix: each POD coordinate received one independent circular shift, and the same shifted realization was then sliced for all eight origins. This construction preserves the overlap-induced dependence between origin statistics under the null. It corrects the invalid alternative of shifting each overlapping origin independently.

All eight origin-level plus-one lower-tail p -values were between 0.135 and 0.830; none was at most 0.05. The predeclared global statistic was the median quadrature residual across origins:

$$T_{\text{obs}} = 0.662734.$$

Twenty-five of 199 synchronized null medians were at or below the observed value, giving

$$p = \frac{25 + 1}{199 + 1} = 0.13.$$

The diagnostic is exploratory and is not adjusted for choosing W, d, r from the same development outcomes. In a spontaneous stochastic trajectory, the statistic also contains dynamical variance as well as projection error [9]. It therefore does not verify a mode family.

11 Can the fitted modes reconstruct the transformed activity?

Prediction remained the selection criterion, but a claimed mode family should at least describe the transformed activity used to fit it. Three development candidates were therefore audited in-sample using:

1. reconstruction R^2 in the complete all-neuron norm, including variance discarded by the POD basis; and
2. overlap between normalized aggregate POD power spectra.

Both diagnostics are optimistic because the same fitting window determines the modes and their amplitudes.

The normalized spectral overlap is insensitive to total amplitude and can be moderately high even when the reconstruction explains virtually no all-neuron variance. The 180-s observed aggregate peak was not recovered at any origin. These fitted low-rank modes do not precisely capture the

Table 7: Descriptive training reconstruction and spectral adequacy.

role	W, d, r	median reconstruction R^2	median spectral overlap	peak matches
comparable minimum	180, 1, 12	0.000065	0.798843	0/8
one-SE fallback	180, 1, 2	0.000029	0.641193	0/8
largest local increment	45, 1, 4	0.000277	0.740397	3/8

transformed neural activity, even in-sample. The contrast with the roughly 0.06 projection-oracle ceiling shows that this failure is not solely a consequence of choosing $q = 256$; the fitted low-rank dynamics use almost none of the available representational capacity.

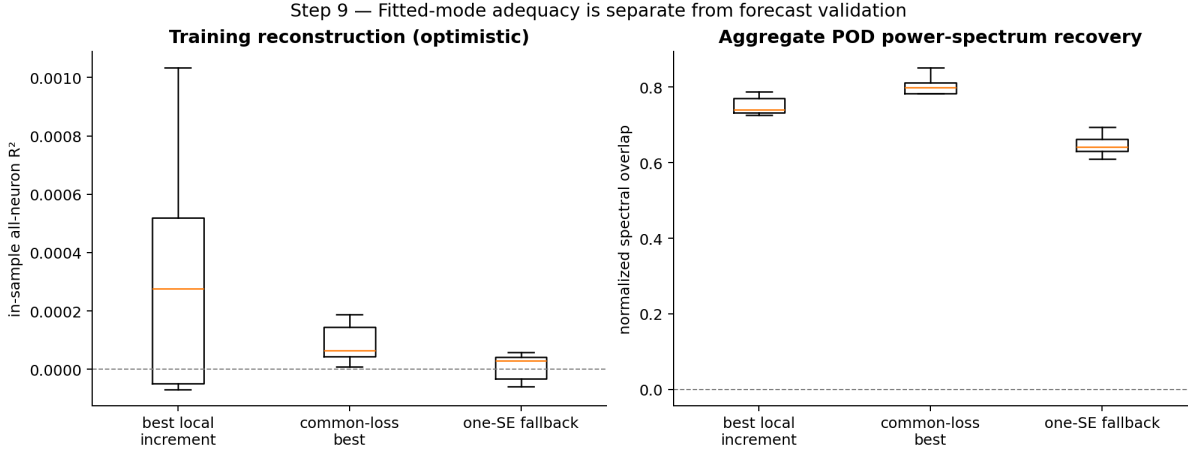


Figure 11: Optimistic in-sample all-neuron reconstruction and normalized spectral overlap. Spectral-shape overlap must not be mistaken for variance capture.

12 Development decision

Table 8: Frozen development decision.

criterion	observed	pass
Positive fixed-reference configuration	0/143	no
Technical multi-reference forecast gate	0/143	no
Practical local $R^2 \geq 0.05$	best 0.008932	no
Gate-passing delete-pair stability	no passing pool	no
Damping-qualified pair in the fallback	none	no
Low-overlap recurrent exact-Hankel track	none verified	no
Global synchronized residual diagnostic	$p = 0.13$	no
Raw-rate practical sensitivity	best 0.010190	no
Convergence-clean optimized-DMD fit	0/128	no

The development result is **not satisfactory for scientific mode tracking**. The negative outcome is not an implementation failure: the pipeline completed, numerical identities passed, candidate failures and warnings were disclosed, and the observation-space target was attainable. Rather,

the present observable and estimator did not produce useful common-reference prediction, stable selection, reconstructive capture, or verified recurrent modes.

The two window statements must remain distinct:

- $W = 180$ s minimizes the comparable forecast loss, but that loss is still worse than the fixed acquisition mean. The rank-2 one-SE result is a fallback only.
- $W = 45$ s maximizes a small increment beyond its own inferior local mean, but it has strongly negative common-reference performance and no damping-qualified oscillatory pair.

Neither window is verified. Opening the outer tail to choose another window, delay, rank, smoother, or mode would spend confirmation data on continued development, so its score files intentionally contain headers only.

13 Recommended standard pipeline and next experiment

The verification yields the following standard procedure:

1. define state- and acquisition-safe observables in physical units;
2. learn scaling and any fixed spatial basis on separate data;
3. declare one prediction reference that is constant across all windows, and report window-specific local increments separately;
4. audit whether the observation space can attain the declared effect size under both references;
5. tune W, d, r jointly with an explicit transitions-per-rank guard and a common-rank slice that isolates W, d ;
6. use common chronological origins, endpoint-initialized autonomous multi-horizon forecasts, and fixed-mean, local-mean, persistence, and AR baselines;
7. disclose conditioning, rank-boundary support, every warning, and every fit failure rather than treating a returned object as a successful fit;
8. require optimized estimators to converge before comparing their scores;
9. audit in-sample all-neuron reconstruction and spectral shape separately;
10. require frequency resolution, fitted-component salience, and an explicit damping/lifetime criterion before calling an eigenpair a recurrent oscillation;
11. establish recurrence across low-overlap windows, preferably separated bouts or animals, rather than adjacent highly overlapping contexts;
12. generate one synchronized surrogate realization across all overlapping origins and either fix the candidate in advance or repeat selection inside each surrogate;
13. freeze preprocessing and hyperparameters before opening an outer bout or mouse exactly once.

The next useful experiment is not a wider W, d, r search on these same targets. It is a small prespecified observable comparison using separated NREM bouts or animals. Reasonable candidates are a few causal rate filters with biologically interpretable time constants and the fluorescence observable. If a compact planning comparison is required, the 180-s, $d = 1, r = 2$ fallback and the 45-s, $d = 1, r = 4$ local-increment stress test can be carried forward as explicitly unverified contrasts. Neither should be promoted. Sliding-window Grassmann tracking should begin only after one local estimator passes prediction, reconstruction, damping, and recurrence checks on separated data.

Reproducibility

The executable tutorial is `DMD/scripts/02_systematic_window_tuning_mouse02_sleep.py`. The completed run uses PyDMD 2025.8.1 and seed 20260717. Complete origin and horizon scores, oracle tables, conditioning and warning tables, mode inventories and matches, synchronized surrogate residuals, reconstruction diagnostics, figures, the frozen decision, and a run manifest are in `DMD/results/06_window_parameter_tuning/`. The manifest records the exact script hash used for each completed run. The implementation contains noise-free autonomous-forecast, independent-pair G, A, L , and exact all-neuron POD-score identity checks. The broader DMD numerical test suite contains 25 unit tests.

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